

## Demonstration of Proposed Features

|                      |                                                             |
|----------------------|-------------------------------------------------------------|
| <b>Date</b>          | 27 June 2026                                                |
| <b>Team ID</b>       | XXXXXX                                                      |
| <b>Project Name</b>  | OptiCrop: Smart Agricultural Production Optimization Engine |
| <b>Maximum Marks</b> | 1 Mark                                                      |

### Demonstration of Proposed Features:

This document captures all the features proposed during the planning phase of the OptiCrop project and tracks their implementation and demonstration status. It provides evidence of the successful development and deployment of the agricultural production optimization system.

| S.No | Feature Name                       | Description                                                                              | Status<br>(Implemented<br>/ Partial /<br>Pending) | Demonst<br>rate d<br>(Yes /<br>No) | Remarks                          |
|------|------------------------------------|------------------------------------------------------------------------------------------|---------------------------------------------------|------------------------------------|----------------------------------|
| 1    | Smart Crop Recommendation          | Recommends the most suitable crop based on soil nutrients and environmental conditions   | Implemented                                       | Yes                                | Successfully tested              |
| 2    | Crop Suitability Assessment        | Evaluates whether environmental conditions are suitable for a selected crop              | Implemented                                       | Yes                                | Accurate predictions obtained    |
| 3    | Environmental Parameter Analysis   | Processes Nitrogen, Phosphorous, Potassium, temperature, humidity, pH, and rainfall data | Implemented                                       | Yes                                | Integrated with prediction model |
| 4    | Machine Learning Prediction Engine | Uses ML algorithms for crop recommendation and optimization                              | Implemented                                       | Yes                                | Random Forest model selected     |
| 5    | Web-Based User Interface           | Provides user-friendly input and output pages using Flask                                | Implemented                                       | Yes                                | Fully functional                 |
| 6    | Agricultural Research Analysis     | Supports analysis of crop-environment relationships through visualizations               | Partial                                           | Yes                                | Additional features planned      |

**Feature Implementation Summary:**

|                                 |        |
|---------------------------------|--------|
| <b>Total Features Proposed</b>  | 6      |
| Total Features Implemented      | 5      |
| Total Features Demonstrated     | 6      |
| Overall Implementation Rate (%) | 83.33% |

The OptiCrop project successfully implemented and demonstrated most of the proposed features. The system effectively provides crop recommendations, evaluates environmental suitability, and supports agricultural research through machine learning-based analysis. Future enhancements will focus on expanding research analytics and integrating real-time environmental data sources.